Does prior averaging help when neural recordings are limited?

Two independent simulation studies with known connectivity

Known connectivity
Uncertain physiology

→

Limited recordings
Fixed recording budget

→

Stimulus response
Prespecified target

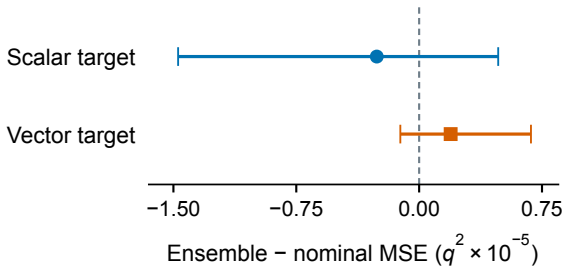
Broad matched uncertainty · M = 3 · endpoints excluded

A Recording decisions

40 networks · two prespecified comparisons

Different recording sets; actual response MSE

97.5% paired network-cluster intervals



Scalar: −0.259 [−1.471, 0.483]

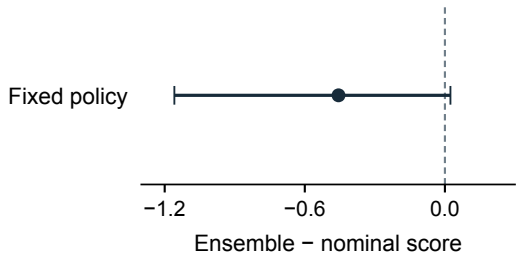
Vector: 0.192 [−0.115, 0.683]

B Risk forecasts

40 new networks · one prespecified comparison

Same data and point predictions

95% paired network-cluster interval



Score: −0.456 [−1.159, 0.024]

MSE-forecast score: log(v) + e²/v

Primary comparisons did not establish a consistent ensemble advantage.

Negative differences favor the ensemble in both panels.

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